7.1 Polygon Angle Sums

Every convex polygon's interior angles can be found by splitting the polygon into triangles from one vertex: an n-gon splits into $(n-2)$ triangles, and since each triangle contributes $180°$, the interior angle sum is $S = (n-2)\cdot 180°$. Exterior angles (one per vertex, formed by extending each side) always add to exactly $360°$ for *any* convex polygon, no matter how many sides it has — walking once around the perimeter turns you through one full rotation. For a **regular** polygon (congruent sides and angles), split these totals evenly: each interior angle is $\frac{(n-2)180°}{n}$, and each exterior angle is $\frac{360°}{n}$. At any single vertex, the interior and exterior angles are supplementary. This worksheet goes beyond plugging into a formula — expect triangulation reasoning, two-variable algebra, and word problems that combine both formulas.

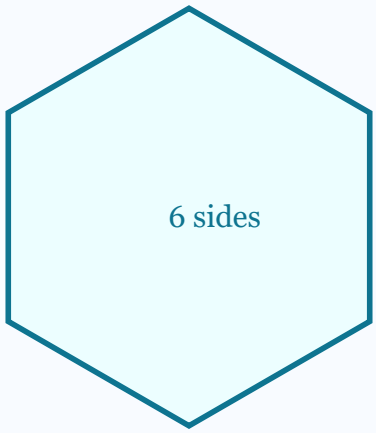
Examples

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Example 1: Triangulate a hexagon (see why the formula works)

Find the sum of the interior angles of a convex hexagon by triangulating it from one vertex, then confirm with the formula.

Solution



A hexagon splits into $6 - 2 = 4$ triangles when diagonals are drawn from one vertex.

Draw diagonals from one vertex to every non-adjacent vertex. This splits the hexagon into $6 - 2 = 4$ triangles.

Each triangle's angles sum to $180°$, and together the triangles' angles make up exactly the hexagon's interior angles, so:

$S = 4 \cdot 180° = 720°$.

Check with the formula: $S = (n - 2)180° = (6 - 2)180° = 720°$. ✓

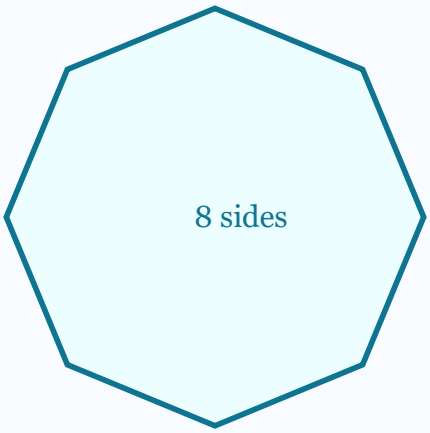
Answer: $720°$.

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Example 2: Regular octagon — two methods

A stop sign is a regular octagon. Find the measure of each interior angle two different ways.

Solution



A regular octagon: all 8 sides and all 8 interior angles are congruent.

Method 1 (interior sum): $S = (8-2)180° = 1080°$. Divide evenly among the 8 congruent angles: $\frac{1080°}{8} = 135°$.

Method 2 (exterior angle): Each exterior angle is $\frac{360°}{8} = 45°$. Since interior and exterior angles at a vertex are supplementary, interior $= 180° - 45° = 135°$.

Both methods agree.

Answer: $135°$.

Example 3: Find n from the interior sum

The interior angles of a convex polygon sum to $1440°$. How many sides does it have?

Solution

Set up the formula and solve for n:

$(n-2)180° = 1440°$

$n - 2 = \dfrac{1440}{180} = 8$

$n = 10.$

Check: $(10-2)180° = 8 \cdot 180° = 1440°$. ✓

Answer: a decagon (10 sides).

Example 4: Regular 15-gon — exterior then interior

A regular polygon has 15 sides. Find each exterior angle, then each interior angle, and verify your interior angle with the sum formula.

Solution

Exterior angle: $\dfrac{360°}{15} = 24°$.

Interior angle (supplementary to exterior): $180° - 24° = 156°$.

Verify: interior sum should be $(15-2)180° = 2340°$; and $15 \cdot 156° = 2340°$. ✓

Answer: exterior $= 24°$, interior $= 156°$.

Example 5: Missing angle in an irregular pentagon

A pentagon has angles $100°$, $110°$, $95°$, $120°$, and x. Find x.

Solution

Interior sum for a pentagon: $S = (5-2)180° = 540°$.

Add the known angles: $100 + 110 + 95 + 120 = 425°$.

$x = 540° - 425° = 115°$.

Answer: $x = 115°$.

Example 6: Does the exterior-sum rule hold for a triangle?

Verify that the exterior angles of an equilateral triangle sum to $360°$, just like any other convex polygon.

Solution

An equilateral triangle has three congruent interior angles of $60°$ each.

Each exterior angle is supplementary to its interior angle: $180° - 60° = 120°$.

Sum of the three exterior angles: $3 \cdot 120° = 360°$.

So the exterior-angle-sum rule ($360°$) holds true even for the smallest possible polygon, $n = 3$.

This rule never depends on n.

Example 7: Find n from an exterior angle

Each exterior angle of a regular polygon measures $40°$. How many sides does the polygon have?

Solution

The exterior angles of a regular n-gon are all equal and sum to $360°$, so each one is $\dfrac{360°}{n}$.

$$40° = \frac{360°}{n} \Rightarrow n = \frac{360°}{40°} = 9.$$

Answer: a regular nonagon (9 sides).

Example 8: Algebraic quadrilateral

A quadrilateral has angles $3x$, $2x$, $4x$, and x. Find x and each angle, then check the sum.

Solution

Interior sum of a quadrilateral: $(4-2)180° = 360°$.

$3x + 2x + 4x + x = 360° \Rightarrow 10x = 360° \Rightarrow x = 36°$.

Angles: $3x = 108°$, $2x = 72°$, $4x = 144°$, $x = 36°$.

Check: $108 + 72 + 144 + 36 = 360°$. ✓

Example 9: Nonagon interior sum

Find the interior angle sum of a convex nonagon (9 sides), and the measure of each angle if it is regular.

Solution

$S = (9-2)180^\circ = 7 \cdot 180^\circ = 1260^\circ.$

If regular, each angle $= \dfrac{1260^\circ}{9} = 140^\circ.$

Answer: sum $= 1260^\circ$; each angle (if regular) $= 140^\circ$.

Example 10: Interior sum as a multiple of the exterior sum

A convex polygon's interior angle sum is exactly 3 times its exterior angle sum. Find the number of sides.

Solution

The exterior sum is always 360° for a convex polygon, so the interior sum here is $3 \cdot 360^\circ = 1080^\circ$.

$(n-2)180^\circ = 1080^\circ$

$n - 2 = 6$

$n = 8.$

Answer: 8 sides (an octagon).

Example 11: Can regular pentagons tile a floor?

A contractor wants to tile a floor edge-to-edge using only regular pentagon tiles, with tiles meeting perfectly at every point (angles around a point must add to exactly 360°). Determine whether this is possible.

Solution

Each interior angle of a regular pentagon: $\dfrac{(5-2)180^\circ}{5} = \dfrac{540^\circ}{5} = 108^\circ.$

For tiles to meet perfectly at a point with no gaps or overlaps, 108° must divide evenly into 360°:

$\dfrac{360^\circ}{108^\circ} = 3.\overline{3}$ — not a whole number.

Answer: No. Three pentagons leave a $360 - 3(108) = 36^\circ$ gap, and a fourth would overlap. (This is why floor and wall tiles are usually squares, equilateral triangles, or regular hexagons — the only regular polygons whose interior angle divides evenly into 360°.)

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Interior sum

Find the interior angle sum of a convex heptagon ($n = 7$).

Answer: **Verification:**

Problem 2: Regular polygon

Find each interior angle of a regular decagon.

Answer: **Verification:**

Problem 3: Find n

The interior angle sum is $2340°$. Find n.

Answer: **Verification:**

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Problem 4: Exterior angle

Find each exterior angle of a regular 20-gon.

Answer: **Verification:**

Problem 5: Missing angle

A hexagon has five angles of 120° each; find the sixth.

Answer: **Verification:**

Problem 6: Exterior sum

Find the exterior angle sum of a convex 100-gon.

Answer: **Verification:**

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Problem 7: Find n

A regular polygon has each exterior angle $24°$. Find n.

Answer: **Verification:**

Problem 8: Regular polygon

Find each interior angle of a regular pentagon.

Answer: **Verification:**

Problem 9: Algebra

A quadrilateral has angles $2x$, $3x$, $4x$, x. Find the largest angle.

Answer: **Verification:**

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Problem 10: Interior sum

Find the interior angle sum of a convex dodecagon ($n = 12$).

Answer: **Verification:**

Problem 11: Interior from exterior

One exterior angle is $30°$. Find the interior angle at that same vertex.

Answer: **Verification:**

Problem 12: Find n

A regular polygon has each interior angle $150°$. Find n.

Answer: **Verification:**

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Problem 13: Missing angle

A pentagon has angles $90°$, $100°$, $110°$, $120°$, and x. Find x.

Answer: **Verification:**

Problem 14: Challenge

A regular polygon's interior angle is exactly 8 times its exterior angle. Find the number of sides.

Answer: **Verification:**

Problem 15: Challenge

A convex nonagon has interior angles that are nine consecutive integers (in degrees). Find the smallest and largest angles.

Answer: **Verification:**

✓ Answer Key

1. 900°
2. 144°
3. $n = 15$
4. 18°
5. 120°
6. 360°
7. $n = 15$
8. 108°
9. 144° ($x = 36°$)
10. 1800°
11. 150°
12. $n = 12$
13. $x = 120°$
14. $n = 18$ (each exterior = 20°, each interior = 160°)
15. Smallest 136°, largest 144° (sum = 1260°)

Great work — keep practising!